

Sep 15 - Intro to Alignment Algorithms: Global and Local

Sequence comparison

- Biomolecular sequences
 - DNA - string over 4 letter alphabet {A, C, G, T}
 - RNA - string over 4 letter alphabet {A, C, G, U}
 - Protein - string over 20 letter alphabets (amino acids)
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Global similarity

- **Input** - two sequences over the same alphabet
- **Output** - alignment of the two sequences
 - **Match** - same letters; suggests same ancestor
 - **Mismatch** - different letters
 - **Indel** - gap represented by dash; can be cluster of multiple dashes; insertion or deletion
- **Scoring scheme** - tells you value of match, mismatch, or indel
 - Want to find out maximum score alignment
 - **Unit-score** - common scoring scheme that gives highest score for identical sequences
 - If letters match, δ is 1 and otherwise 0 (0 if both -)
 - **Score = sum of pairwise alignment**
 - Ex. $\delta_{(A,A)} + \delta_{(C,G)} + \delta_{(G,G)} = 1 + 0 + 1 = 2$
 - Can add gaps (-)/shifts for optimal alignment given by highest score
 - Example alignment
 - Criticism - does not matter the order of letters
 - **Dayhoff score** - sophisticated scoring scheme for protein sequences
 - Partial alignment for Monkey and Trout somatotropin proteins
- **Scoring functions**
 - Mutations - substitutions, insertions, deletions
 - Scoring function - sum of terms each for a pair of aligned residues and for each gap
 - The meaning - log of the relative likelihood that the sequences are related, compared to being unrelated
 - Positive terms - identities & conservative substitutions
 - Negative terms - nonconservative substitutions
- **Edit graph**
 - Graph where alignments between the two sequences correspond to paths between the begin and end nodes of the graph
 - Algorithm has three lines
 - Example - sequence AGC and AC
 - Edit graph has $(3+1)*(2+1)$ nodes
 - Any path from (0,0) to (2,3) corresponds to alignment, which corresponds to path
 - Each directed edge (shown by pink oval) has labelled pair of aligned letters

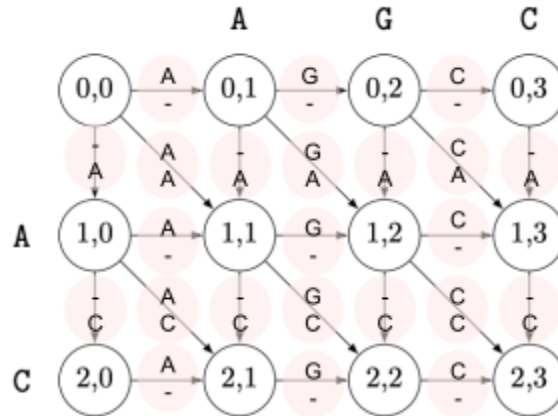


Figure 2. Edit Graph for AC and AGC.

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■ Optimal alignment is ↘, →, ↘

- **Principle of optimality**

- Optimal answer to a problem is expressed in terms of optimal answer for its sub problems

- **Dynamic programming**

- Given two sequences X and Y and scoring scheme δ
- Find an optimal alignment of X with Y (max score alignment)
 - Can have several answers
- Part 1 - compute first optimal alignment score
- Part 2 - construct optimal alignment
 - Look for optimal alignment = maximal score path in edit graph for begin to end vertex

- Matrix $S = [S(i,j)]$

- $S(i,j)$ = score of maximal cost path from begin vertex and vertex (i,j)

- Optimal path to (i,j) must pass through one of the vertices:

- $(i-1, j)$
 - $(i, j-1)$
 - $(i-1, j-1)$

- If entering (i,j) horizontally:

- $S(i-1, j) + \delta(x_i, -)$

- If entering (i,j) diagonally:

- $S(i-1, j-1) + \delta(x_i, y_j)$

- If entering (i,j) vertically:

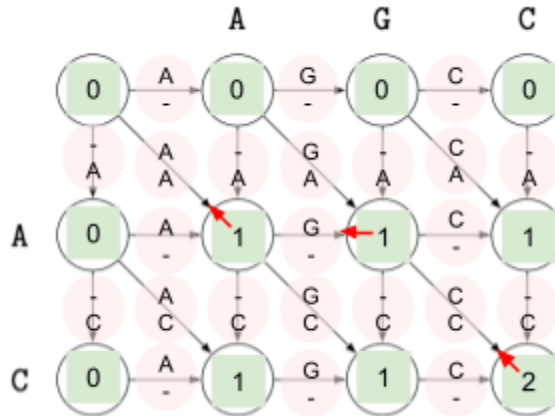
- $S(i, j-1) + \delta(-, y_j)$

$$S(i,j) = \text{MAX} \begin{cases} S(i-1, j-1) + \delta(x_i, y_j), \\ S(i-1, j) + \delta(x_i, -), \\ S(i, j-1) + \delta(-, y_j), \end{cases}$$

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- **Traceback**

- Initialize first row and first column with 0s



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- Optimal alignment AGC over A-C

Local similarity

- Much more difficult than global alignment
 - In global, maximum score is rightmost element of matrix

$$S(i,j) = \text{MAX} \begin{cases} 0, \\ S(i-1, j-1) + \delta(x_i, y_j), \\ S(i-1, j) + \delta(x_i, -), \\ S(i, j-1) + \delta(-, y_j), \end{cases}$$

General scoring schemes

- Assumptions
 - Independence of mutations at different sites
 - Additive scoring scheme
 - Gaps of any length are considered one mutation
 - All of the efficient alignment algorithms, employing on the dynamic programming method, are based fundamentally on the fact that the scoring function is additive

Heuristic interpretation of the alignment score

X = AACGTAA

Y = ACGGTGT

Alignment *

A A C G - T A A

| | | |

A - C G G T G T

- 4 matches, 2 single letter indels, 2 mismatches

Let p be the probability of a match

Let q be the probability of a substitution/mismatch

Let r be the probability of a single letter indel/gap

What is the probability of this alignment?

$$P_{\text{align}(*)} = p^4 q^2 r^2$$

The log-likelihood is:

$$s' = \log P_{\text{align}(*)} = 4 \log p + 2 \log q + 2 \log r$$

$$\begin{aligned} s = s' - 8 \log k &= 4 \log p + 2 \log q + 2 \log r - 8 \log k \\ &= 4 \log p + 2 \log q + 2 \log r - (4 \log k + 2 \log k + 2 \log k) \\ &= 4 \log (p/k) + 2 \log (q/k) + 2 \log (r/k) \\ &= \mathbf{4 - 2\mu - 2\gamma} \end{aligned}$$

choose k so $\log (p/k) = 1$

$\mu = \log (k/q)$ = penalty for mismatch

$\gamma = \log (k/r)$ penalty for gap